PASADHI · SOLUTION PAPER

PASADHI: one pool from the machines you already own

PASADHI contributors · v1.0

23 October 2026 · MIT for the open core

github.com/camponggogo/pasadhi · pasadhi.tech

Abstract

Organisations already own the hardware they need; what they lack is a way to use it as one machine. PASADHI turns office PCs, servers, laptops and small GPUs into a single compute pool with four properties: it is **matrix-first**, **layered as processes**, **priced by measured impact**, and **honest about the physics**. On a 50-machine test bed we measure 15.7 req/s from one gateway worker and 22.3–24.8 req/s from two; a content addressed prefix cache lifts time-to-first-token on an 8 000-token document from 111.4 s to 0.71 s (157×); a document-aware router reaches 90.15 % accuracy in 31.5 µs; and a mixed fleet of 1 450 cards (2 GB to 40 GB) aggregates 9.54 PFLOPS f32 and 47.7 POPS int8 at 143.5 kW. The design rule that survives every measurement is simple: **the inner loop stays local, and the price follows the measurement**.

1. The problem, stated exactly

A cloud GPU is rented by the hour and priced per token. An office already owns machines that idle for sixteen hours a day. The naive arithmetic — sum the RAM, sum the FLOPs — fails for a reason that has nothing to do with silicon: coordination cost. Our first 50-machine run made this visible in one table: throughput sat at 12 req/s while latency grew linearly, and at 50 concurrent clients **96.5 % of the response time was queueing** above a 174 ms service floor. The fleet was not slow; the control path was.

2. The solution in one picture

```
 L5 storage     content-addressed blocks, local disks
L4 kernel      GEMM · int8 GEMM · FFT/conv as GEMM
L3 scheduler   tiling, placement, back-pressure, retries
L2 policy      identity, scopes, budgets, audit
L1 transport   framed TCP; shared memory on the same host
```

Each layer is a process group with its own bounded queue. A slow layer blocks its producer, which is the signal the scheduler needs; no layer can stall another. The gateway and the node plane are separate processes, so a heartbeat storm cannot delay an inference request.

3. What we measured, and what it bought

Change	Before	After
Node plane separated from the gateway	44 req/s heartbeat, 210 ms at C=1	**161 req/s per shard, 105 ms**
Live node list cached for 5 s	4.9 req/s at C=8	**15.0 req/s**
Two gateway workers	15.7 req/s	**22.3 / 24.8 req/s** (C=8 / C=16)
Prefix cache, 8 000-token document	111.4 s to first token	**0.71 s, 88.2 % reuse**
Document-aware router	—	**90.15 %, 31.5 µs**
Domain experts instead of a mixture	86 machines, infeasible	**32 machines, 0.64 req/s (10.7×)**
Mixed fleet of 1 450 cards	—	**9.54 PFLOPS f32, 47.7 POPS, 143.5 kW**

Three numbers deserve emphasis because they decide architecture, not tuning:

1. **The gateway ceiling is the event loop.** A raw health endpoint serves 58.7–71.2 req/s; a completion costs ≈60 ms of loop time. Workers scale until the host does: four workers peak at 29.1 req/s at C=8 and fall to 20.5 at C=32.
2. **Prefix reuse is the largest single win.** 157× on the median, with p95 unchanged — the cache fixes the median, not the tail.
3. **Ports are not throughput.** Two ports beat two workers by 16 % because each process owns its accept queue; a userspace proxy gives the gain back (−25 %), and a hundred ports buy 4.4 GiB of memory and no extra request per second.

4. Capacity, priced by impact

Contributions are paid by what they actually contribute, weighted from the same sweep that produced the performance numbers: **CPU 98.0 %**, RAM 0.4 %, disk 1.0 %, network 0.6 % on a CPU-only node; **CPU 56.6 %, GPU 42.2 %** once a GTX 1050 is present. A four-thread machine shared at 100 % earns exactly 2.50 THB per hour; an eight-thread machine at 35 % earns ≈711 THB per month.

That honesty changes the commercial model. At a 30 % share a 150-machine office earns 53 460 THB per month against a 149 745 THB subscription, so the break-even share is ≈60 %, or the group price must fall to ≈350 THB per machine. The product is therefore sold three ways: **share and use free**, **rent monthly/yearly**, or **buy an on-prem licence** scoped to a group, unlimited machines, departments rolled up, multi-licence merge for enterprises.

5. The fleet we would build

Tier	What	Why
Office tier	100–200 desktops and laptops, 20–30 % each	the capacity already exists and is paid for
Accelerator tier	small cards (2–16 GB) for independent jobs	22× per card on the same job; independent jobs need no network
Anchor tier	a few 24–80 GB cards	the only place a 35 GB job can run; without them the tail owns the clock
Network	1 GbE for the office tier, 10 GbE only for the anchor tier	sharded GEMM keeps 6.9 % of compute across an antipodal fibre link and 32.3 % over LEO; the inner loop must stay local

6. What is deliberately not in the product

- **No 50-way pipeline sharding.** S^* is 3–10 stages, overhead is 150 ms per step, and activation traffic would be 65.5 Mbit/s — 1 922× the measured gateway traffic.
- **No interplanetary inner loop.** A $4\,096^3$ GEMM keeps 0.004 % of compute across Earth–Mars; only store-and-forward, checkpointed work crosses that link.
- **No claim of cheaper FLOPs than a GPU.** A used-card fleet lands at ฿0.0010 per delivered TFLOP, the same as a rented A100. The margin comes from software, support and residency, not from silicon.

7. Roadmap

Milestone	Evidence it is done
M1 packed int8 kernel in C/AVX2	25–45 GFLOP/s on the reference part, ≤1.5 % error
M2 shared-memory transport	pipeline ≥1.5× the single loop at 512^2 blocks
M2.7 group controller	join tokens, quotas, offline autonomy, per-department billing
M3 distributed matrix plane	row-panel sharding, retry-safe tiles, recovery after a mid-panel kill
M6 public network	redundant execution 2× + 5 % spot checks, credit ledger with a daily Merkle root

8. Limits

Every number here comes from one host (8 vCPU, Docker Desktop), a mock backend and published card specifications. The fleet ceilings move with the host; the shape of the results does not. Where a figure is modelled rather than measured, the text says so.

Appendix: where each number lives

`report/RPS_LIMIT.md` (gateway ceiling), `report/MULTIPORT_100.md` (ports), `report/KV_CACHE.md` (prefix reuse), `report/EXPERTS_WHITEPAPER.md` (router), `report/HETERO_FLEET.md` (mixed fleet), `report/IMPACT_PRICING.th.md` (weights), `report/GROUP.th.md` (group economics), `pasadhi/core/orbit.py` (link budgets).